Oral presentation

Foodborne pathogens in milk and milk products along the water buffalo milk chain in Bangladesh

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Shuvo Singha[1,2,3,4], Gerrit Koop[4,5], Md. Mizanur Rahman[3,4], Fabrizio Ceciliani[1,4], Md. Matiar Rahman Howlader[2], Sofia Boqvist[6], **Domenico Vecchio[4,7]**, Paola Cremonesi[8], Md. Nazmul Hoque[9], Ylva Persson[4,10], Cristina Lecchi[1,4]

1: *Department of Veterinary Medicine and Animal Sciences, Università degli Studi di Milano;* **2:** *Department of Physiology, Veterinary, Animal and Biomedical Sciences, Sylhet Agricultural University*; **3:** *Department of Medicine and Surgery, Chattogram Veterinary and Animal Sciences University*; **4:** *Udder Health Bangladesh*; **5:** *Department of Population Health Sciences, Faculty of Veterinary Medicine, Utrecht University*; **6:** *Department of Biomedical Sciences and Veterinary Public Health, The Swedish University of Agricultural Sciences*; **7:** *Italian National Reference Centre on Water Buffalo Farming and Productions Hygiene and Technologies (CReNBuf), Istituto Zooprofilattico Sperimentale del Mezzogiorno*; **8:** *Institute of Agricultural Biology and Biotechnology, National Research Council*; **9:** *Department of Gynecology, Obstetrics and Reproductive Health, Bangabandhu Sheikh Mujibur Rahman Agricultural University*; **10:** *National Veterinary Institute*

Correspondence: Shuvo Singha (shuvo.singha@unimi.it)

Introduction

Milk and milk products are nutritionally rich and are considered to be important components for many healthy diets. However, if contaminated, they can be a source of pathogenic microorganisms. Bacteria are responsible for two-thirds of human foodborne diseases, with a relatively heavy burden affecting low and middle-income countries (LMIC). Although buffalo milk accounts for 35% of the total milk production of Asian countries, there are limited reports estimating the prevalence of foodborne pathogens from buffalo milk in LMIC. The buffalo milk chain in Bangladesh is a supply chain consisting of activities and processes, including production, processing, trading, and consumption of milk and milk products. The farms are typically situated far from the milk processing centers, and hygienic practices in milk handling are minimal. Consequently, when local transportation is prolonged, and labor-intensive measures are required, the absence of milk cooling facilities can compromise milk safety due to potential contamination by foodborne pathogens. Thus, water buffalo milk and milk products may be a source for transmitting foodborne pathogens to humans.

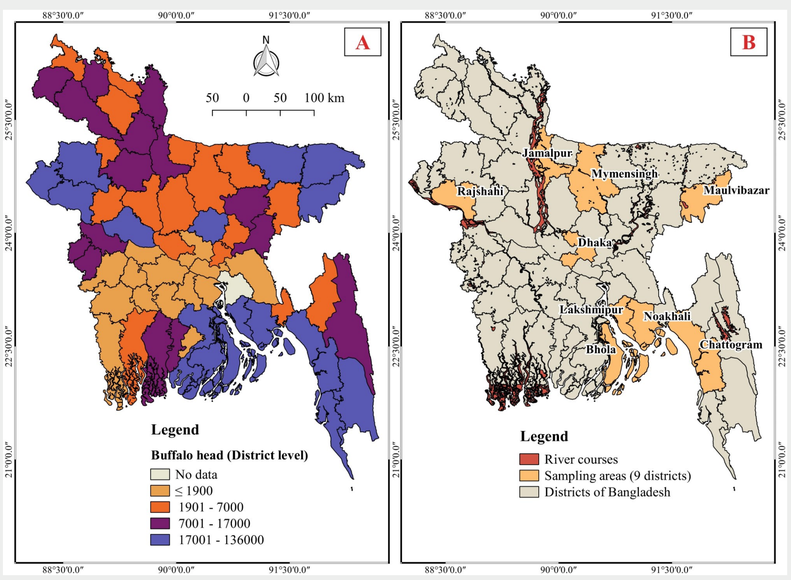
Objectives

Estimate the prevalence of seven important foodborne pathogens

Identify the factors associated with these pathogens in milk and milk products along the buffalo milk chain in Bangladesh

(*S. aureus, E. coli, E. coli O157:H7 eae stx1 stx2, L. monocytogenes, Y. enterocolitica, S. enterica, C. jejuni*)

Materials and Methods



A= *Buffalo population in BD*
B= *Selected districts for sampling*

143 samples were randomly selected

Bulk milk (n=34)

Middleman (n= 37)

Milk collection center (n= 37)

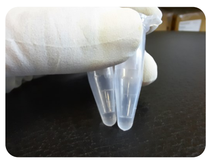
Milk products (n= 35)

DNA extraction from milk and milk products

1 mL of sample

Centrifuge at 16000g for 1min

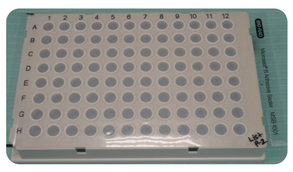
Maxwell 16 cell DNA extractor



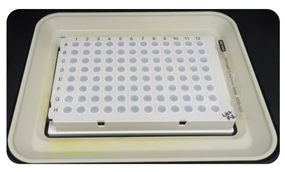
Purified DNA

DNA quantity assessment

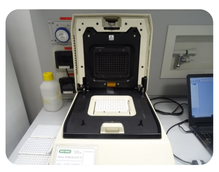
Performing the Real-time PCR *(Positive at < 35 cycles)*



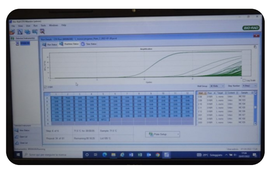
Preparation of PCR plate



Setting the PCR plate

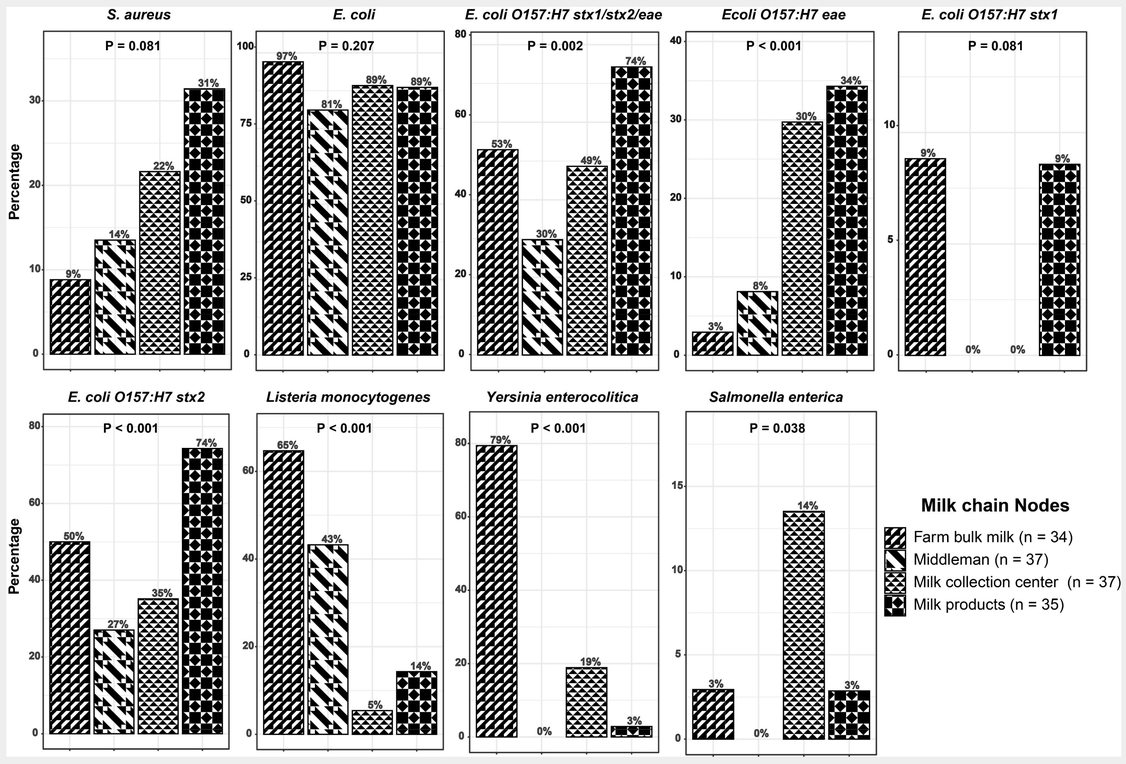


Performing PCR using BIO-RAD RT-PCR



Interpreting PCR results

Results



Farm-level variables associated with the prevalence of zoonotic pathogens (P < 0.15)

Variable name	Categories	N	*E. coli*		*E. coli O157:H7*		*L. monocytogenes*		*Y. enterocolitica*	
			Positive (%)	P	Positive (%)	P	Positive (%)	P	Positive (%)	P
Season	Late autumn	11	11 (100.0)	0.12	3 (27.3)	0.14	5 (45.5)	0.11	11 (100.0)	0.01
	Spring	19	19 (100.0)		12 (63.2)		15 (78.9)		15 (78.9)	
	Winter	4	3 (75.0)		3 (75.0)		2 (50.0)		1 (25.0)	
District	Noakhali	11			6 (54.5)	< 0.01	9 (81.8)	0.08	9 (81.8)	0.02
	Bhola	5			5 (100.0)		4 (80.0)		5 (100.0)	
	Moulvibazar	4			3 (75.0)		2 (50.0)		1 (25.0)	
	Mymensingh	4			3 (75.0)		0 (0.0)		4 (100.0)	
	Jamalpur	6			0 (0.0)		4 (66.7)		6 (100.0)	
	Rajshahi	4			1 (0.25)		3 (75.0)		2 (50.0)	
Rearing system	Free range	26			16 (61.5)	0.11				
	Intensive	8			2 (25.0)					
Keep bulk milk containers open	Yes	30			18 (60.0)	0.08				
	No	3			0 (0.0)					
Previous history of clinical mastitis in last 12 months	Yes	31			31 (100.0)	0.09				
	No	3			2 (66.7)					

Middleman, milk collection center, and product shops related variables associated with the prevalence of zoonotic pathogens (P < 0.15)

Variable name	Categories	N	*S. aureus*		*E. coli*		*E. coli O157:H7*		*Y. enterocolitica*	
			Positive (%)	P	Positive (%)	P	Positive (%)	P	Positive (%)	P
District[a]	Noakhali	20			19 (0.0)	0.01				
	Bhola	5			4 (80.0)					
	Moulvibazar	4			3 (75.0)					
	Mymensingh	3			3 (100.0)					
	Jamalpur	4			1 (25.0)					
	Rajshahi	1			0 (0.0)					
Season[a]	Late autumn	7			4 (57.1)	0.06				
	Spring	6			4 (66.7)					
	Summer	20			19 (95.0)					
	Winter	4			3 (75.0)					
Type of trader[a]	Farmer or middleman	18			12 (66.7)	0.09				
	Wholesalers	16			15 (93.8)					
How the container is dried[a]	Sundry	15			14 (93.3)	0.17	1 (6.7)	0.01		
	Cloth or tissue	16			11 (68.8)		9 (56.3)			
									a Middleman level risk factors;	
Type of trader[b]	Farmer or middleman	20								
	Wholesalers	17								
Nature of milk[b]	Buffalo milk	25							1 (4.0)	< 0.01
	Buffalo-cow mixed milk	11							6 (54.5)	
									b Milk collection centers level risk factors;	
Type of container[c]	Earthen	10	2 (20.0)	0.04						
	Plastic	18	9 (50.0)							
	Glass	7	0 (0.0)							
									c Milk products level risk factors	

Conclusion

- Most of the foodborne pathogen prevalence was very high
- Prevalence of *E. coli* O157:H7 and *S. aureus*, increased over the milk value chain
- On farms, *Y. enterocolitica* and *L. monocytogenes* were the most prevalent
- Using to same cloth for wiping milk containers should be avoided
- Use of glass or stainless steel was more helpful than plastic-made containers

Acknowledgements

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